CHAPTER 1

Definitions

The successful implementation of various investment strategies necessitates a sound working knowledge of the fundamentals of options and option trading. The option strategist must be familiar with a wide range of the basic aspects of stock options—how the price of an option behaves under certain conditions or how the markets function. A thorough understanding of the rudiments and of the strategies helps the investor who is not familiar with options to decide not only whether a strategy seems desirable, but also—and more important—*whether it is suitable*. Determining suitability is nothing new to stock market investors, for stocks themselves are not suitable for every investor. For example, if the investor's primary objectives are income and safety of principal, then bonds, rather than stocks, would be more suitable. The need to assess the suitability of options is especially important: Option buyers can lose their entire investment in a short time, and uncovered option writers may be subjected to large financial risks. Despite follow-up methods designed to limit risk, the individual investor must decide whether option trading is suitable for his or her financial situation and investment objective.

ELEMENTARY DEFINITIONS

A stock option is the right to buy or sell a particular stock at a certain price for a limited period of time. The stock in question is called the underlying security. A call option gives the owner (or holder) the right to buy the underlying security, while a put option gives the holder the right to sell the underlying security. The price at which the stock may be bought or sold is the exercise price, also called the striking price. (In the listed options market, "exercise price" and "striking price" are synonymous.) A stock option affords this right to buy or sell for only a limited period of time; thus, each option has an expiration date. Throughout the book, the term "options" is always understood to mean listed options, that is, options traded on national option exchanges where a secondary market exists. Unless specifically mentioned, over-the-counter options are not included in any discussion.

DESCRIBING OPTIONS

Four specifications uniquely describe any option contract:

1. the type (put or call),
2. the underlying stock name,
3. the expiration date, and
4. the striking price.

As an example, an option referred to as an "XYZ July 50 call" is an option to buy (a call) 100 shares (normally) of the underlying XYZ stock for $50 per share. The option expires in July. The price of a listed option is quoted on a per-share basis, regardless of how many shares of stock can be bought with the option. Thus, if the price of the XYZ July 50 call is quoted at $5, buying the option would ordinarily cost $500 ($5 × 100 shares), plus commissions.

THE VALUE OF OPTIONS

An option is a "wasting" asset; that is, it has only an initial value that declines (or "wastes" away) as time passes. It may even expire worthless, or the holder may have to exercise it in order to recover some value before expiration. Of course, the holder may sell the option in the listed option market before expiration.

An option is also a security by itself, but it is a derivative security. The option is irrevocably linked to the underlying stock; its price fluctuates as the price of the underlying stock rises or falls. Splits, stock dividends, and special cash dividends affect the terms of listed options, although cash dividends do not. The holder of a call does not receive any cash dividends paid by the underlying stock.

STANDARDIZATION

The listed option exchanges have standardized the terms of option contracts. The terms of an option constitute the collective name that includes all of the four descriptive specifications. While the type (put or call) and the underlying stock are self-evident and essentially standardized, the striking price and expiration date require more explanation.

Striking Price. Striking prices are spaced at different intervals for stocks, ETFs, and indices, depending on the underlying price and the liquidity of the option contracts. With rare exceptions, striking prices are 5 points apart at most, even for high-priced stocks like GOOG and AAPL or indices like SPX (the S&P 500 Index). For lower-priced stocks, strikes are often just 1 point apart, provided that the options on that stock are relatively heavily traded. There are occasionally 2.5-point strikes as well, centered between 5-point striking price intervals. The simplest way to tell what strikes are available on the stock you wish to trade is to look at a quote montage (a free one is available online at www.cboe.com, for example).

Expiration Dates. Options have expiration dates in one of three fixed cycles:

1. the January/April/July/October cycle,
2. the February/May/August/November cycle, or
3. the March/June/September/December cycle.

In addition, the two nearest months have listed options as well. However, at any given time, the longest-term expiration dates are normally no farther away than 9 months. Longer-term options, also called LEAPS (which stands for Long-term Equity Anticipation Securities), are available on some stocks (see Chapter 25). Hence, in any cycle, options may expire in 3 of the 4 major months (series) plus the near-term months. For example, on February 1 of any year, XYZ options may expire in February, March, April, July, and October—not in January. The February option (the closest series) is the *short-* or *near-term* option; and the October, the *far-* or *long-term* option. If there were LEAPS options on this stock, they would expire in January of the following year and in January of the year after that.

The exact date of expiration is fixed within each month. The last trading day for an option is the third Friday in the expiration month. Although the option actually does not expire until the following day (the Saturday following), a public customer must invoke the right to buy or sell stock by notifying his broker by 5:30 p.m., New York time, on the last day of trading.

The above expiration date—the Saturday after the third Friday—is considered the "regular" option expiration date in any given month. But in recent years, options have been introduced that expire at other times. There are *quarterly* options that expire on the last trading day of March, June, September, and December. There are only a few of these—mostly likely on large indices.

There are also *weekly* options. These are generally listed on a Thursday and expire on the Friday that occurs eight calendar days later. (*Note:* These weeklys *do* expire on Friday, not the Saturday following.) An ever-growing number of stocks and indices have weekly listed options, and it is likely that the option exchanges will seek to add even more, since these are popular with the public.

The Price of an Option. Options trade in one-penny increments in many cases. Otherwise, prices are quoted in 5-cent increments. For stock options, the trading increment is one penny up until the price of the option reaches $3.00. After that, the trading increment is 5 cents. However, for liquid Exchange Traded Funds (ETFs) and indices, options can trade in pennies at any price.

THE OPTION ITSELF: OTHER DEFINITIONS

Classes and Series. A class of options refers to all put and call contracts on the same underlying security. For instance, all IBM options—all the puts and calls at various strikes and expiration months—form one class. A series, a subset of a class, consists of all contracts of the same class (IBM, for example) having the same expiration date and striking price.

Opening and Closing Transactions. An *opening transaction* is the initial transaction, either a buy or a sell. For example, an opening buy transaction creates or increases a long position in the customer's account. A *closing transaction* reduces the customer's position. Opening buys are often followed by closing sales; correspondingly, opening sells often precede closing buy trades.

Open Interest. The option exchanges keep track of the number of opening and closing transactions in each option series. This is called the open interest. Each opening transaction adds to the open interest and each closing transaction decreases the open interest. The open interest is expressed in a number of option contracts, so that one order to buy 5 calls opening would increase the open interest by 5. Note that the open interest does not differentiate between buyers and sellers—there is no way to tell if there is a preponderance of either one. While the magnitude of the open interest is not an extremely important piece of data for the investor, it is useful in determining the liquidity of the option in question. If there is a large open interest, then there should be little problem in making fairly large trades. However, if the open interest is small—only a few hundred contracts outstanding—then there might not be a reasonable secondary market in that option series.

The Holder and Writer. Anyone who buys an option as the initial transaction—that is, buys opening—is called the holder. On the other hand, the investor who sells an option as the initial transaction—an opening sale—is called the writer of the option. Commonly, the writer (or seller) of an option is referred to as being short the option contract. The term "writer" dates back to the over-the-counter days, when a direct link existed between buyers and sellers of options; at that time, the seller was the writer of a new contract to buy stock. In the listed option market, however, the issuer of all options is the Options Clearing Corporation, and contracts are standardized. This important difference makes it possible to break the direct link between the buyer and seller, paving the way for the formation of the secondary markets that now exist.

Exercise and Assignment. An option owner (or holder) who invokes the right to buy or sell is said to exercise the option. Call option holders exercise to buy stock; put holders exercise to sell. The holder of most stock options may exercise the option at any time after taking possession of it, up until 8:00 p.m. on the last trading day; the holder does not have to wait until the expiration date itself before exercising. (*Note:* Some options, called "European" exercise options, can be exercised only *on* their expiration date and not before—but they are generally not stock options.) These exercise notices are irrevocable; once generated, they cannot be recalled. In practical terms, they are processed only once a day, after the market closes. Whenever a holder exercises an option, somewhere a writer is assigned the obligation to fulfill the terms of the option contract: Thus, if a call holder exercises the right to buy, a call writer is assigned the obligation to sell; conversely, if a put holder exercises the right to sell, a put writer is assigned the obligation to buy. A more detailed description of the exercise and assignment of call options follows later in this chapter; put option exercise and assignment are discussed later in the book.

RELATIONSHIP OF THE OPTION PRICE AND STOCK PRICE

In- and Out-of-the-Money. Certain terms describe the relationship between the stock price and the option's striking price. A call option is said to be out-of-the-money if the stock is selling below the striking price of the option. A call option is in-the-money if the stock price is above the striking price of the option. (Put options work in a converse manner, which is described later.)

Example: XYZ stock is trading at $47 per share. The XYZ July 50 call option is out-of-the-money, just like the XYZ October 50 call and the XYZ July 60 call. However, the XYZ July 45 call, XYZ October 40, and XYZ January 35 are in-the-money.

The *intrinsic value* of an in-the-money call is the amount by which the stock price exceeds the striking price. If the call is out-of-the-money, its intrinsic value is zero. The price that an option sells for is commonly referred to as the premium. The premium is distinctly different from the time value premium (called time premium, for short), which is the amount by which the option premium itself exceeds its intrinsic value. The time value premium is quickly computed by the following formula for an in-the-money call option:

Call time value premium = Call option price + Striking price – Stock price

Example: XYZ is trading at 48, and XYZ July 45 call is at 4. The premium—the total price—of the option is 4. With XYZ at 48 and the striking price of the option at 45, the in-the-money amount (or intrinsic value) is 3 points (48–45), and the time value is 1 (4–3).

If the call is out-of-the-money, then the premium and the time value premium are the same.

Example: With XYZ at 48 and an XYZ July 50 call selling at 2, both the premium and the time value premium of the call are 2 points. The call has no intrinsic value by itself with the stock price below the striking price.

An option normally has the largest amount of time value premium when the stock price is equal to the striking price. As an option becomes deeply in- or out-of-the-money, the time value premium shrinks substantially. Table 1-1 illustrates this effect. Note that

TABLE 1-1.
Changes in time value premium.

XYZ Stock Price	XYZ Jul 50 Call Price	Intrinsic Value	Time Value Premium
40	½	0	½
43	1	0	1
35	2	0	2
47	3	0	3
→50	**5**	**0**	**5**
53	7	3	4
55	8	5	3
57	9	7	2
60	10.50	10	.50
70	19.50	20	–.50*

*Simplistically, a deeply in-the-money call may actually trade at a discount from intrinsic value, because call buyers are more interested in less expensive calls that might return better percentage profits on an upward move in the stock. This phenomenon is discussed in more detail when arbitrage techniques are examined.

TABLE 1-2.
Comparison of XYZ stock and call prices.

Striking Price	+	XYZ July 45 Call Price	–	XYZ Stock Price	=	Over Parity
(45	+	1	–	45.50)	=	.50
(45	+	2.50	–	47)	=	.50
(45	+	5.50	–	50)	=	.50
(45	+	15.50	–	60)	=	.50

the time value premium increases as the stock nears the striking price (50) and then decreases as it draws away from 50. The term "time value premium" implies that this part of an option's premium is completely due to the time remaining. In fact, that isn't really true. The *volatility* of the underlying stock has a great deal to do with how much "time premium" is in the option. So, really, "time premium" is something of a misnomer, but it's the standard term.

Parity. An option is said to be trading *at parity* with the underlying security if it is trading for its intrinsic value. Thus, if XYZ is 48 and the XYZ July 45 call is selling for 3, the call is *at parity*. A common practice of particular interest to option writers (as shall be seen later) is to refer to the price of an option by relating how close it is to parity with the common stock. Thus, the XYZ July 45 call is said to be a half-point over parity in any of the cases shown in Table 1-2.

FACTORS INFLUENCING THE PRICE OF AN OPTION

An option's price is the result of properties of both the underlying stock and the terms of the option. The major quantifiable factors influencing the price of an option are the:

IMPORTANT

1. price of the underlying stock,
2. striking price of the option itself,
3. time remaining until expiration of the option,
4. volatility of the underlying stock,
5. current risk-free interest rate (such as for 90-day Treasury bills), and
6. dividend rate of the underlying stock.

The first four items are the major determinants of an option's price, while the latter two are generally less important, although the dividend rate can be influential in the case of high-yield stock.

THE FOUR MAJOR DETERMINANTS

Probably the most important influence on the option's price is the stock price, because if the stock price is far above or far below the striking price, the other factors have little influence. Its dominance is obvious on the day that an option expires. On that day, only the stock price and the striking price of the option determine the option's value; the other four factors have no bearing at all. At this time, an option is worth only its intrinsic value.

Example: On the expiration day in July, with no time remaining, an XYZ July 50 call has the value shown in Table 1-3; each value depends on the stock price at the time.

The Call Option Price Curve. The call option price curve is a curve that plots the prices of an option against various stock prices. Figure 1-1 shows the axes needed to graph such a curve. The vertical axis is called Option Price. The horizontal axis is for Stock Price. This figure is a graph of the intrinsic value. When the option is either out-of-the-money or equal to the stock price, the intrinsic value is zero. Once the stock price passes the striking price, it reflects the increase of intrinsic value as the stock price goes up. Since a call is usually worth at least its intrinsic value at any time, the graph thus represents the minimum price that a call may be worth.

When a call has time remaining to its expiration date, its total price consists of its

TABLE 1-3.
XYZ option's values on the expiration day.

XYZ Stock Price	XYZ July 50 Call (Intrinsic) Value at Expiration
40	0
45	0
48	0
50	0
52	2
55	5
60	10

FIGURE 1-1.
The value of an option at expiration, its intrinsic value.

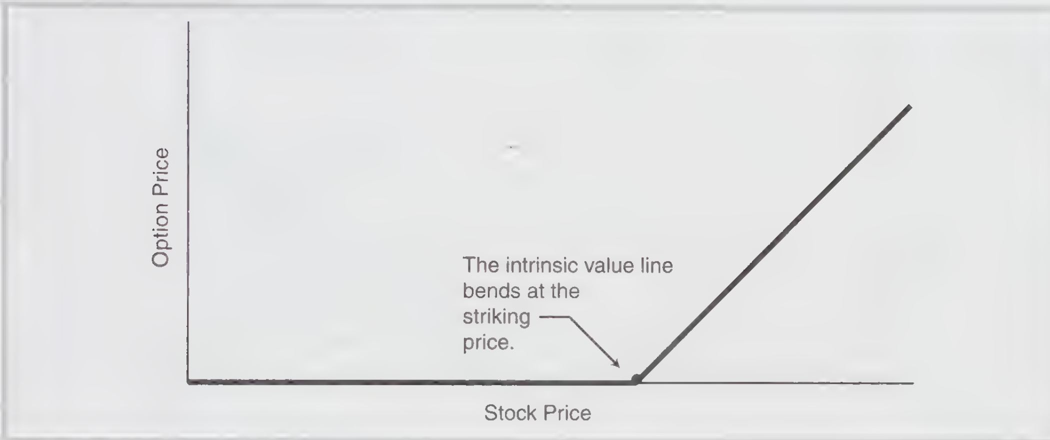


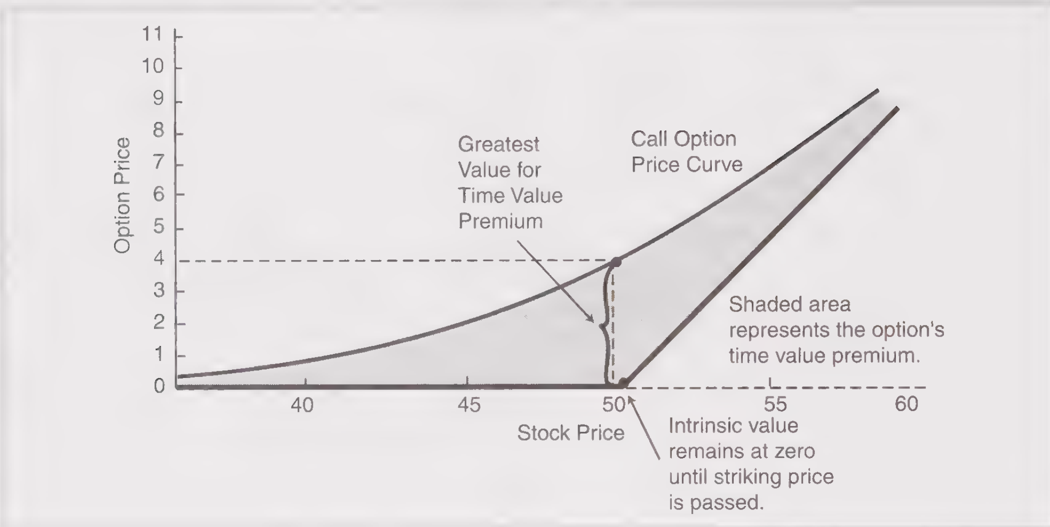
TABLE 1-4.
The prices of a hypothetical July 50 call with 6 months of time remaining, plotted in Figure 1-2.

XYZ Stock Price (Horizontal Axis)	XYZ July 50 Call Price (Vertical Axis)	Intrinsic Value	Time Value Premium (Shading)
40	1	0	1
45	2	0	2
48	3	0	3
→50	**4**	**0**	**4**
52	5	2	3
55	6.50	5	1.50
60	11	10	1

intrinsic value plus its time value premium. The resultant call option price curve takes the form of an inverted arch that stretches along the stock price axis. If one plots the data from Table 1-4 on the grid supplied in Figure 1-2, the curve assumes two characteristics:

1. The time value premium (the shaded area) is greatest when the stock price and the striking price are the same.

FIGURE 1-2.
Six-month July call option (see Table 1-4).



2. When the stock price is far above or far below the striking price (near the ends of the curve), the option sells for nearly its intrinsic value. As a result, the curve nearly touches the intrinsic value line at either end. [Figure 1-2 thus shows both the intrinsic value and the option price curve.]

This curve, however, shows only how one might expect the XYZ July 50 call prices to behave with 6 months remaining until expiration. As the time to expiration grows shorter, the arched line drops lower and lower, until, on the final day in the life of the option, it merges completely with the intrinsic value line. In other words, the call is worth only its intrinsic value at expiration. Examine Figure 1-3, which depicts three separate XYZ calls. At any given stock price (a fixed point on the stock price scale), the longest-term call sells for the highest price and the nearest-term call sells for the lowest price. At the striking price, the actual differences in the three option prices are the greatest. Near either end of the scale, the three curves are much closer together, indicating that the actual price differences from one option to another are small. For a given stock price, therefore, option prices decrease as the expiration date approaches.

Example: On January 1st, XYZ is selling at 48. An XYZ July 50 call will sell for more than an April 50 call, which in turn will sell for more than a January 50 call.

This statement is true no matter what the stock price is. The only reservation is that with the stock deeply in- or out-of-the-money, the actual difference between the

FIGURE 1-3.
Price Curves for the 3-, 6-, and 9-month call options.

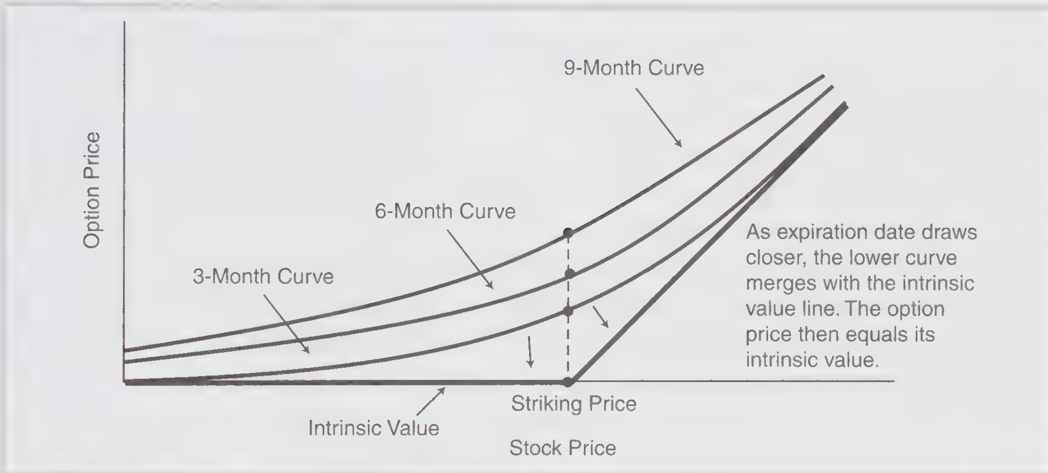
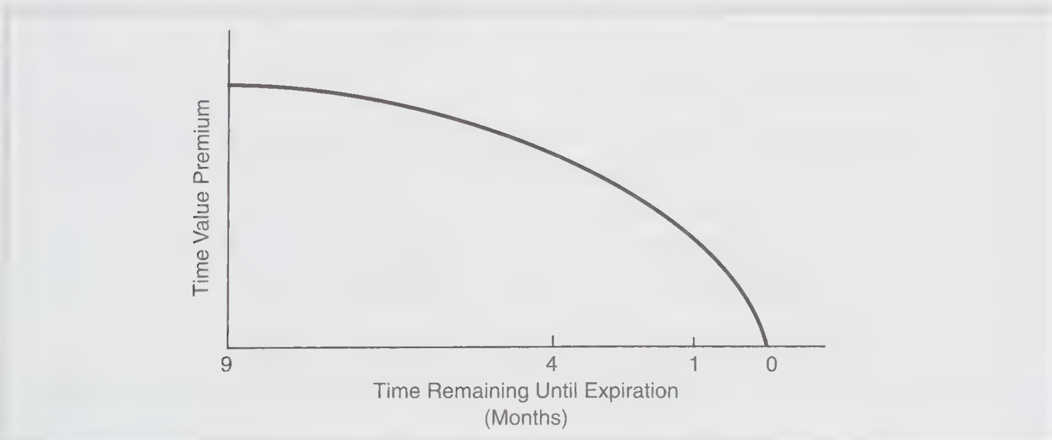


FIGURE 1-4.
Time value premium decay, assuming the stock price remains constant.



January, April, and July calls will be smaller than with XYZ stock selling at the striking price of 50.

Time Value Premium Decay. In Figure 1-3, notice that the price of the 9-month call is not three times that of the 3-month call. Note next that the curve in Figure 1-4 for the decay of time value premium is not straight; that is, *the rate of decay of an option is not linear.* An option's time value premium decays much more rapidly in the last few weeks of its life (that is, in the weeks immediately preceding expiration) than it does in the first few weeks of its existence. The rate of decay is actually related to the square root of the time remaining. Thus, a 3-month option decays (loses time value premium) at twice the rate of a 9-month option, since the square root of 9 is 3. Similarly, a 2-month option decays at twice the rate of a 4-month option ($\sqrt{4} = 2$).

This graphic simplification should not lead one to believe that a 9-month option necessarily sells for twice the price of a 3-month option, because the other factors also influence the actual price relationship between the two calls. Of those other factors, the *volatility* of the underlying stock is particularly influential. *More volatile underlying stocks have higher option prices.* This relationship is logical, because if a stock has the ability to move a relatively large distance upward, buyers of the calls are willing to pay higher prices for the calls—and sellers demand them as well. For example, if AT&T and Xerox sell for the same price (as they have been known to do), the Xerox calls would be more highly priced than the AT&T calls because Xerox is a more volatile stock than AT&T.

The interplay of the four major variables—stock price, striking price, time, and volatility—can be quite complex. While a rising stock price (for example) is directing the price of a call upward, decreasing time may be simultaneously driving the price in the opposite direction. Thus, the purchaser of an out-of-the-money call may wind up with a loss even after a rise in price by the underlying stock, because time has eroded the call value.

THE TWO MINOR DETERMINANTS

The Risk-Free Interest Rate. This rate is generally construed as the current rate of 90-day Treasury bills. Higher interest rates imply slightly higher option premiums, while lower rates imply lower premiums. Although members of the financial community disagree as to the extent that interest rates actually affect option price, they remain a factor in most mathematical models used for pricing options. (These models are covered much later in this book.)

The Cash Dividend Rate of the Underlying Stock. Though not classified as a major determinant in option prices, this rate can be especially important to the writer (seller) of an option. If the underlying stock pays no dividends at all, then a call option's worth is strictly a function of the other five items. *Dividends, however, tend to lower call option premiums: The larger the dividend of the underlying common stock, the lower the price of its call options.* One of the most influential factors in keeping option premiums low on high-yielding stock is the yield itself.

Example: XYZ is a relatively low-priced stock with low volatility selling for \$25 per share. It pays a large annual dividend of \$2 per share in four quarterly payments of \$.50 each. What is a fair price of an XYZ call with striking price 25?

A prospective buyer of XYZ options is determined to figure out a fair price. In six months XYZ will pay $1 per share in dividends, and the stock price will thus be reduced by $1 per share when it goes ex-dividend over that time period. In that case, if XYZ's price remains unchanged except for the ex-dividend reductions, it will then be $24. Moreover, since XYZ is a nonvolatile stock, it may not readily climb back to 25 after the ex-dividend reductions. Therefore, the call buyer makes a low bid — even for a 6-month call—because the underlying stock's price will be reduced by the ex-dividend reduction, and the call holder does not receive the cash dividends.

This particular call buyer calculated the value of the XYZ July 25 call in terms of what it was worth with the stock discounted to 24—not at 25. He knew for certain that the stock was going to lose 1 point of value over the next 6 months, provided the dividend rate of XYZ stock did not change. In actual practice, option buyers tend to discount the upcoming dividends of the stock when they bid for the calls. However, not all dividends are discounted fully; usually the nearest dividend is discounted more heavily than are dividends to be paid at a later date. The less-volatile stocks with the higher dividend payout rates have lower call prices than volatile stocks with low payouts. In fact, in certain cases, an impending large dividend payment can substantially increase the probability of an exercise of the call in advance of expiration. (This phenomenon is discussed more fully in the following section.) In any case, to one degree or another, dividends exert an important influence on the price of some calls.

OTHER INFLUENCES

These six factors, major and minor, are only the quantifiable influences on the price of an option. In practice, nonquantitative market dynamics—investor sentiment—can play various roles as well. In a bullish market, call premiums often expand because of increased demand. In bearish markets, call premiums may shrink due to increased supply or diminished demand. These influences, however, are normally short-lived and generally come into play only in dynamic market periods when emotions are running high.

EXERCISE AND ASSIGNMENT: THE MECHANICS

The holder of an option can exercise his right at any time during the life of an option: Call option holders exercise to buy stock, while put option holders exercise to sell stock. In the event that an option is exercised, the writer of an option with the same terms is *assigned an obligation* to fulfill the terms of the option contract.

EXERCISING THE OPTION

The actual mechanics of exercise and assignment are fairly simple, due to the role of the Options Clearing Corporation (OCC). As the issuer of all listed option contracts, it controls all listed option exercises and assignments. Its activities are best explained by an example.

Example: The holder of an XYZ January 45 call option wishes to exercise his right to buy XYZ stock at $45 per share. He instructs his broker to do so. The broker then notifies the administrative section of the brokerage firm that handles such matters. The firm then notifies the OCC that they wish to exercise one contract of the XYZ January 45 call series.

Now the OCC takes over the handling. OCC records indicate which member (brokerage) firms are short or which have written and not yet covered XYZ Jan 45 calls. The OCC selects, at random, a member firm that is short at least one XYZ Jan 45 call, and it notifies the short firm that it has been assigned. That firm must then deliver 100 shares of XYZ at $45 per share to the firm that exercised the option. The assigned firm, in turn, selects one of its customers who is short the XYZ January 45 call. This selection for the assignment may be either:

1. at random,
2. on a first-in/first-out basis, or
3. on any other basis that is fair, equitable, and approved by the appropriate exchange.

The selection of the customer who is short the XYZ January 45 completes the exercise/assignment process. (If one is an option writer, he should obviously determine exactly how his brokerage firm assigns its option contracts.)

HONORING THE ASSIGNMENT

The assigned customer *must* deliver the stock—he has no other choice. It is too late to try buying the option back in the option market. He must, without fail, deliver 100 shares of XYZ stock at $45 per share. The assigned writer does, however, have a choice as to how to fulfill the assignment. If he happens to be already long 100 shares of XYZ in his account, he merely delivers that 100 shares as fulfillment of the assignment notice. Alternatively, he can go into the stock market and buy XYZ at the current market price—presumably something higher than $45—and then deliver the newly purchased stock as fulfillment. A third alternative is merely to notify his brokerage firm that he wishes to go short XYZ stock and to ask them to deliver the 100 shares of XYZ at 45 out of his short account. At times, borrowing stock to go short may not be possible, so this third alternative is not always available on every stock.

Margin Requirements. If the assigned writer purchases stock to fulfill a contract, reduced margin requirements generally apply to the transaction, so that he would not have to fully margin the purchased stock merely for the purpose of delivery. Generally, the customer only has to pay a day-trade margin of the difference between the current price of XYZ and the delivery price of $45 per share. If he goes short to honor the assignment, then he has to fully margin the short sale at the current rate for stock sold short on a margin basis.

AFTER EXERCISING THE OPTION

The OCC and the customer exercising the option are not concerned with the actual method in which the delivery is handled by the assigned customer. They want only to ensure that the 100 shares of XYZ at 45 are, in fact, delivered. The holder who exercised the call can keep the stock in his account if he wants to, but he has to margin it fully or pay cash in a cash account. On the other hand, he may want to sell the stock immediately in the open market, presumably at a higher price than 45. If he has an established margin account, he may sell right away without putting out any money. If he exercises in a cash account, however, the stock must be paid for in full—even if it is subsequently sold on the same day. Alternatively, he may use the delivered stock to cover a short sale in his own account if he happens to be short XYZ stock.

COMMISSIONS

Both the buyer of the stock via the exercise and the seller of the stock via the assignment are charged a full stock commission on 100 shares, unless a special agreement exists between the customer and the brokerage firm. Generally, option holders incur higher commission costs through assignment than they do selling the option in the secondary market. *So the public customer who holds an option is better off selling the option in the secondary market than exercising the call.*

Of course, sometimes a customer wants to own the underlying stock, perhaps because it is attractive to him or because he wants to cover a short sale. In these cases, the exercise of the call would be desirable.

ANTICIPATING ASSIGNMENT

The writer of a call often prefers to buy the option back in the secondary market, rather than fulfill the obligation via a stock transaction. It should be stressed again that once the writer receives an assignment notice, it is too late to attempt to buy back (cover) the call. The writer must buy *before* assignment, or live up to the terms upon assignment. The writer who is aware of the circumstances that generally cause the holders to exercise can anticipate assignment with a fair amount of certainty. In anticipation of the assignment, the writer can then close the contract in the secondary market. As long as the writer covers the position at any time during a trading day, he cannot be assigned on that option. Assignment notices are determined on open positions as of the *close* of trading each day. The crucial question then becomes, "How can the writer anticipate assignment?" Several circumstances signal assignments:

1. a call that is in-the-money at expiration,
2. an option trading at a discount prior to expiration, or
3. the underlying stock paying a large dividend and about to go ex-dividend.

Automatic Exercise. Assignment is all but certain if the option is in-the-money at expiration. Should the stock close even a penny above the striking price on the last day of trading, the holder will normally exercise. In fact, even if the call trades in-the-money at any time during the last trading day, it might be exercised if a market-maker has taken an offsetting position and issued an exercise notice. Even if an option holder forgets that he owns an option and fails to exercise, the OCC automatically exercises any option that is one penny in the money at expiration, unless the customer—through his brokerage firm—gives specific instructions not to exercise. This *automatic exercise* mechanism ensures that no investor throws money away through carelessness.

Example: XYZ closes at 50 on the third Friday of January, the last trading day for the January option series. Since options don't expire until Saturday, the next day, the OCC and all the brokerage firms have the opportunity to review their records to see if any options should have been profitably exercised but were not. If a customer owned a January 45 call, but failed to either sell or exercise it (perhaps due to illness or an inability to get to a phone or computer while traveling), it is automatically exercised. Since it is worth $500 per contract, less the commissions for buying the stock, the customer stands to receive a substantial amount of money back. However, he is now long the stock, so that must be sold to realize the gain.

In the above example, note that the exercise creates a stock purchase in the account. If there is not sufficient cash in the account to make this purchase, a margin call may result. This is why some brokerage firms do not allow option buying in retirement accounts (IRAs, for example), because one cannot just routinely add money into an IRA to fulfill the requirements of an automatic exercise.

Any writer who wishes to avoid an assignment notice should always buy back (or cover) the option if it appears that the stock will be above the strike at expiration. The probability of being assigned is virtually 100% if the option expires in the money, even by a penny.

Early Exercise Due to Discount. When options are exercised *prior to expiration*, this is called *early*, or *premature*, *exercise*. The writer can usually expect an early exercise when the call is trading at or below parity. A parity or discount situation in advance of expiration may mean that an early exercise is forthcoming, even if the discount is slight. A writer who does not want to deliver stock should buy back the option prior to expiration if the option is apparently going to trade at a discount to parity. The reason is that arbitrageurs (floor traders or member firm traders who pay only minimal commissions) can take advantage of discount situations. (Arbitrage is discussed in more detail later in the text; it is mentioned here to show why early exercise often occurs in a discount situation.)

Example: XYZ is bid at $50 per share, and an XYZ January 40 call option is offered at a discount price of 9.80. The call is actually "worth" 10 points. The arbitrageur can take advantage of this situation through the following actions, all on the same day:

1. Buy the January 40 call at 9.80.
2. Sell short XYZ common stock at 50.
3. Exercise the call to buy XYZ at 40.

The arbitrageur makes 10 points from the short sale of XYZ (steps 2 and 3), from which he deducts the 9.80 points he paid for the call. Thus, his total gain is 20 cents—the amount of the discount. Since he pays only a minimal commission, this transaction results in a net profit.

Also, if the writer can expect assignment when the option has no time value premium left in it, then conversely the option will usually not be called if time premium is left in it.

Example: Prior to the expiration date, XYZ is trading at 50½, and the January 50 call is trading at 1. The call is not necessarily in imminent danger of being called, since it still has half a point of time premium left.

$$\begin{aligned} \text{Time value premium} &= \text{Call price} + \text{Striking price} - \text{Stock price} \\ &= 1 + 50 - 50.50 \\ &= .50 \end{aligned}$$

Early Exercise Due to Dividends on the Underlying Stock. Sometimes the market conditions create a discount situation, and sometimes a large dividend gives rise to a discount. Since the stock price is almost invariably reduced by the amount of the dividend, the option price is also most likely reduced after the ex-dividend. Since the holder of a listed option does not receive the dividend, he may decide to sell the option in the secondary market before the ex-date in anticipation of the drop in price. If enough calls are sold because of the impending ex-dividend reduction, the option may come to parity or even to a discount. Once again, the arbitrageurs may move in to take advantage of the situation by buying these calls and exercising them.

If assigned prior to the ex-date, the writer does not receive the dividend for he no longer owns the stock on the ex-date. Furthermore, if he receives an assignment notice on the ex-date, he must deliver the stock with the dividend. It is therefore very important for the writer to watch for discount situations on the day prior to the ex-date.

A word of caution: Do not conclude from this discussion that a call will be exercised for the dividend if the dividend is larger than the remaining time premium. It won't. An example will show why.

Example: XYZ stock, at 50, is going to pay its regular $1 dividend with the ex-date set for the next day. An XYZ January 40 call is selling at 10.25; it has twenty-five cents of time premium. (TVP = 10.25 + 40 – 50 = .25.) The same type of arbitrage will not work. Suppose that the arbitrageur buys the call at 10.25 and exercises it: He now owns the stock for the ex-date, and he plans to sell the stock immediately at the opening on the ex-date, the next day. On the ex-date, XYZ opens at 49, because it goes ex-dividend by $1. The arbitrageur's transactions thus consist of:

1. Buy the XYZ January 40 call at 10.25.
2. Exercise the call the same day to buy XYZ at 40.
3. On the ex-date, sell XYZ at 49 and collect the $1 dividend.

He makes 9 points on the stock (steps 2 and 3), and he receives a 1-point dividend, for a total cash inflow of 10 points. However, he loses 10.25 points paying for the call. The overall transaction is a loser and the arbitrageur would thus not attempt it.

A dividend payment that exceeds the time premium in the call, therefore, does not imply that the writer will be assigned.

More of a possibility, but a much less certain one, is that the arbitrageur may attempt a "risk arbitrage" in such a situation. *Risk arbitrage* is arbitrage in which the arbitrageur runs the risk of a loss in order to try for a profit. The arbitrageur may suspect that the stock will not be discounted the full ex-dividend amount or that the call's time premium will increase after the ex-date. In either case (or both), he might make a profit: If the stock opens down only 60 cents or if the option premium expands by 40 cents, the arbitrageur could profit on the opening. In general, however, arbitrageurs do not like to take risks and therefore avoid this type of situation. So the probability of assignment as the result of a dividend payment on the underlying stock is small, unless the call trades at parity or at a discount.

Of course, the anticipation of an early exercise assumes rational behavior on the part of the call holder. If time premium is left in the call, the holder is always better off financially to sell that call in the secondary market rather than to exercise it. However, the terms of the call contract give a call holder the right to go ahead and exercise it anyway—even if exercise is not the profitable thing to do. In such a case, a writer would receive an assignment notice quite unexpectedly. Financially unsound early exercises do happen, though not often, and an option writer must realize that, in a very small percentage of cases, he could be assigned under very illogical circumstances.

THE OPTION MARKETS

The trader of stocks does not have to become very familiar with the details of the way the stock market works in order to make money. Stocks don't expire, nor can an investor be pulled out of his investment unexpectedly. However, the option trader is required to do more homework regarding the operation of the option markets. In fact, the option strategist who does not know the details of the working of the option markets will likely find that he or she eventually loses some money due to ignorance.

MARKET-MAKERS

In at least one respect, stock and listed option markets are similar. Stock markets use specialists to do two things: First, they are required to make a market in a stock by buying and selling from their own inventory, when public orders to buy or sell the stock are absent. Second, they keep the public book of orders, consisting of limit orders to buy and sell, as well as stop orders placed by the public. When listed option trading began, the Chicago Board Options Exchange (CBOE) introduced a similar method of trading, the market-maker and the board broker system. The CBOE assigns several market-makers to each optionable stock to provide bids and offers to buy and sell options in the absence of public orders. Market-makers cannot handle public orders; they buy and sell for their own accounts only. A separate person, the board broker, keeps the book of limit orders. The board broker, who cannot do any trading, opens the book for traders to see how many orders to buy and sell are placed nearest to the current market (consisting of the highest bid and lowest offer). (The specialist on the stock exchange keeps a more closed book; he is not required to formally disclose the sizes and prices of the public orders.)

In theory, the CBOE system is more efficient than the stock exchange system. With several market-makers competing to create the market in a particular security, the market should be a more efficient one than a single specialist can provide. Also, the somewhat open book of public orders should provide a more orderly market. In practice, whether the CBOE has a more efficient market is usually a subject for heated discussion. The strategist need not be concerned with the question.

The American Stock Exchange uses specialists for its option trading, but it also has floor traders who function similarly to market-makers. The regional option exchanges use combinations of the two systems; some use market-makers, while others use specialists.

OPTION SYMBOLOGY

From the time that options were listed in 1973 until a complete overhaul of the symbology in 2010, option symbols were a rather arcane combination of letters (to represent strike and expiration month) and 3-character base symbols (even for stocks with four characters in their own symbol, like AAPL). But as more and more stocks, ETFs, and other entities came to have listed options, and these options extended farther out in time (more than one year), the old symbology system became unworkable. It lasted far longer than it should have, mostly due to certain large brokerage firms' and quote vendors' reluctance to change. But eventually, the Options Clearing Corporation enlisted the aid of several industry representatives to create the Options Symbol Initiative (OSI), whose purpose it was to create new option codes. The implementation of the OSI took place in 2010, and the industry is far better for it.

If you want a historic look at option symbology, find a copy of the fourth edition of this book and read the section in the first chapter. It is now a rather amusing look at history, but while it was happening, it took an army of clerks at various firms around the world to keep the option databases updated daily with all of the changes that took place, which included, but were not limited to, splits, special dividends, spinoffs, long-term options (LEAPS), weekly and monthly option expirations, and a myriad of strikes caused by highly volatile movement by the underlying stock, etc.

Today, the situation is simpler, although it is no longer standardized. The OSI suggested how option symbols should be created, but each firm, vendor, and exchange is free to format them how it sees fit. However, the individual customer really doesn't have to worry about this too much, for most of the time brokerage and vendor software displays an option as something like:

IBM Jan (21) 2011 45 call

Where IBM is the stock symbol, Jan is the expiration month, 21 is the expiration day, 2011 is the expiration year, 45 is the striking price, and "call" is the type of option (as opposed to "put").

Some vendors don't show the expiration day if it is the "regular" Saturday following the third Friday. But the day must be shown for weekly or quarterly options. Some vendors only show the last two digits of the expiration year. Some still use the old letter codes for the expiration month ("A" = January call; "B" = February call; . . . "L" = December call; "M" = January put; "N" = February put; . . . "X" = December put).

But in any case, it's a lot more logical now than it used to be, because *numbers* are used for strikes and expiration dates rather than the letters that were used before. And because of the use of numbers for these items, only one stock symbol is necessary for *all* of the options on a particular stock.

DETAILS OF OPTION TRADING

The facts that the strategist should be concerned with are included in this section. They are not presented in any particular order of importance, and this list is not necessarily complete. Many more details are given in the discussion of specific strategies throughout this text.

1. *Option trading can be frenetic near the end of the day.* Listed stock options trade from 9:30 a.m. to 4 p.m. Eastern time. Some index options trade until 4:15 p.m. Eastern time. Waiting too long to place an order at the end of the trading day is not advisable. In the crush of orders that might occur, even a market order could conceivably not be filled. Also, one should be mindful of trading at the end of the last trading day of an option's life (the third Friday for "regular" expiration cycle options, or any Friday for weekly options). Again, it is advisable to place orders with plenty of time to spare, rather than waiting for the "crush" of trading that comes at the end of the option's life.
2. *Option trades have a one-day settlement cycle.* The trade settles on the next business day after the trade. Purchases must be paid for in full, and the credits from sales "hit" the account on the settlement day. Some brokerage firms require settlement on the same day as the trade, when the trade occurs on the last trading day of an expiration series.
3. *Options are opened for trading in rotation.* When the underlying stock opens for trading on any exchange, regional or national, the options on that stock then go into opening rotation on the corresponding option exchange. The rotation system also applies if the underlying stock halts trading and then reopens during a trading day; options on that stock reopen via a rotation.

   In the rotation itself, interested parties make bids and offers for each particular option series one at a time—the XYZ January 45 call, the XYZ January 50 call, and so on—until all the puts and calls at various expiration dates and striking prices have been opened. Trades do not necessarily have to take place in each series, just bids and offers. Orders such as spreads, which involve more than one option, are not executed during a rotation.

While the rotation is taking place, it is possible that the underlying stock could make a substantial move. This can result in option prices that seem unrealistic when viewed from the perspective of each option's opening. Consequently, the *opening* price of an option can be a somewhat suspicious statistic, since none of them open at exactly the same time.

Also, it should be noted that most option traders do *not* trade during rotation, so a market order may receive a very poor price. Hence, if one is considering trading during rotation, a *limit order* should be used. (Order entry is discussed in more detail in a later section of this chapter.)

4. *When the underlying stock splits or pays a stock dividend, or pays a special cash dividend of greater than 12.5 cents, or pays a regular dividend in an amount greater than 10% of the stock price, then the terms of its options are adjusted.* Such an adjustment may result in fractional striking prices and in options for other than 100 shares per contract. No adjustments in terms are made for cash dividends that are regularly declared and paid on a quaterly or other basis. The actual details of splits, stock dividends, and rights offerings, along with their effects on the option terms, are always published by the option exchange that trades those options. Notices are sent to all member firms, who then make that information available to their brokers for distribution to clients. In actual practice, the option strategist should ascertain from the broker the specific terms of the new option series, in case the broker has overlooked the information sent.

Example 1: XYZ is a $50 stock with option striking prices of 45, 50, and 60 for the January, April, and July series. It declares a 2-for-1 stock split. Usually, in a 2-for-1 split situation, the number of outstanding option contracts is doubled and the striking prices are halved. The owner of 5 XYZ January 60 calls becomes the owner of 10 XYZ January 30 calls. Each call is still for 100 shares of the underlying stock.

After the split, XYZ has options with strikes of 22½, 25, and 30. In some cases, the option exchange officials may introduce another strike if they feel such a strike is necessary; in this example, they might introduce a striking price of 20.

Example 2: UVW is trading at 40 with strikes of 35, 40, and 45 for the January, April, and July series. UVW declares a 5% stock dividend. The contractually standardized 100 shares per contract is adjusted up to 105, and the striking prices are reduced by dividing each by 1.05, rounded to the nearest penny. Hence the new strikes are 33.33, 38.10, and

42.86, respectively. In addition, there will be a new base symbol assigned to these "non-100 share" contracts: UVW1 (or UVW2, if UVW1 is already in use).

After the split, the exchange usually opens for trading new strikes of 35, 40, and 45—each for 100 shares of the underlying stock. For a while, there are six striking prices for UVW options. As time passes, the fractional strikes are eliminated as they expire. Since they are not reintroduced, they eventually disappear as long as UVW does not issue further stock dividends.

Example 3: WWW Corp. (symbol WWW) is trading at $120 per share, with strike prices of 110, 115, 120, 125, and 130. WWW declares a 3-for-1 split. In this case, the strike prices would be divided by 3 (and rounded to the nearest penny, becoming 36.67, 38.33, 40, 41.67, and 43.33); the number of contracts in every account would be tripled; and each option would still be an option on 100 shares of WWW stock. The general rule of thumb is that when a split results in round lots (2-for-1, 3-for-1, 4-for-1, etc.), the number of option contracts is increased and the strike price is decreased, and each option still represents 100 shares of the underlying stock.

Example 4: When a split does not result in a round lot, a different adjustment must be used for the options. Suppose that AAA Corp. (symbol: AAA) is trading at $60 per share and declares a 3-for-2 split. In this case, each option's strike will be multiplied by two-thirds (to reflect the 3-for-2 split), *but the number of contracts held in an account will remain the same and each option will be an option on 150 shares of AAA stock.*

Suppose that there were strikes of 55, 60, and 65 preceding this split. After the split, AAA common itself would be trading at $40 per share, reflecting the post-split 3-for-2 adjustment from its previous price of 60. There would be new options with strikes of 36.625, 40, and 43.375 (these had been the pre-split strikes of 55, 60, and 65).

Since each of these options would be for 150 shares of the underlying stock, the exchange creates a *new option base symbol* for these options, because they no longer represent 100 shares of AAA common. Suppose the exchange says that the post-split, 150-share option contracts will henceforth use the option symbol AAA1.

After the split, the exchange will then list "normal" 100-share options on AAA, perhaps with strike prices of 35, 40, and 45. This creates a situation that can sometimes be confusing for traders and can lead to problems. There will actually be two options with striking prices of 40—one for 100 shares and the other for 150 shares. Suppose the July contract is being considered. The option with symbol AAA is a July 40 option on 100 shares of AAA Corp., while the option with symbol AAA1 is a July 40 option on 150 shares of AAA Corp. Since option prices are quoted on a per-share basis, *they will have nearly* *identical price quotes on any quote system* (see item 5). If one is not careful, you might trade the wrong one, thereby incurring a risk that you did not intend to take.

For example, suppose that you sell, as an opening transaction, the AAA1 July 40 call at a price of 3. Furthermore, suppose that you did not realize that you were selling the 150-share option; this was a mistake, but you don't yet realize it. A couple of days later, you see that this option is now selling at 13—a loss of 10 points. You might think that you had just lost $1,000, but upon examining your brokerage statement (or confirms, or day trading sheet), you suddenly see that the loss is $1,500 on that contract! Quite a difference, especially if multiple contracts are involved. This could come as a shock if you thought you were hedged (perhaps you bought 100 shares of AAA common when you sold this call), only to find out later that you didn't have a correctly hedged position in place after all.

Even more severe problems can arise if this stock splits *again* during the life of this option. Sometimes the options will be adjusted so that they represent a non-standard number of shares, such as the 150-share options involved here; and after multiple splits, the exchange may even apply a multiplier to the option, rather than adjusting its strike price repeatedly. This type of thing wouldn't happen on the first stock split, but it might occur on subsequent stock splits, spaced closely together over the life of an option. In such a case, the dollar value of the option moves *much* faster than one would expect from looking at a quote of the contract.

So you must be sure that you are trading the exact contract you intend to, and not relying on the fact that the striking price is correct and the option price quote seems to be in line. One must verify that the option being bought or sold is exactly the one intended. In general, it is a good idea, after a split or similar adjustment, to establish opening positions solely with the standard contracts and to leave the split-adjusted contracts alone.

5. *All options are quoted on a per-share basis,* regardless of how many shares of stock the option involves. Normally the quote assumes 100 shares of the underlying stock. Suppose UVW had paid a 5% stock dividend, so that the options are for 105 shares each. If the UVW April 38.10 is offered at a price of $3.00, it actually costs $315 ($3.00 × 105).
6. *Changes in the price of the underlying stock can bring about new striking prices.* As a stock, index, or ETF moves up and down, it can eventually outdistance the existing striking prices. In that case, the exchange will list new ones. This can even be done at the request of customers, at times. The idea is to have plenty of strikes to choose from at all times. Sometimes, a stock will gap up or down so much that it is suddenly trading in an area where *no* strikes exist at all. Often, the exchange will attempt to get new strikes listed immediately, but occasionally it takes until the next day.

POSITION LIMIT AND EXERCISE LIMIT

7. *An investor or a group of investors cannot be long or short more than a set limit of contracts in one stock on the same side of the market.* The actual limit varies according to the trading activity in the underlying stock. The most heavily traded stocks with a large number of shares outstanding have position limits of 250,000 contracts. Smaller stocks have position limits ranging from as low as 5,000 contracts to 200,000 contracts, depending on liquidity. These limits can be expected to increase over time, if stocks' capitalizations continue to increase. The exchange on which the option is listed makes available a list of the position limits on each of its optionable stocks. So, if one were long the limit of XYZ call options, he cannot at the same time be short any XYZ put options. Long calls and short puts are on the same side of the market; that is, both are bullish positions. Similarly, long puts and short calls are both on the bearish side of the market. While these position limits generally exceed by far any position that an individual investor normally attains, the limits apply to "related" accounts. For instance, a money manager or investment advisor who is managing many accounts cannot exceed the limit when all the accounts' positions are combined.
8. *The number of contracts that can be exercised in a particular period of time (usually 5 business days) is also limited to the same amount as the position limit.* This exercise limit prevents an investor or group from "cornering" a stock by repeatedly buying calls one day and exercising them the next, day after day. *Option exchanges set exact limits, which are subject to change.*

ORDER ENTRY

ORDER INFORMATION

Of the various types of orders, each specifies:

1. whether the transaction is a buy or sell,
2. the option to be bought or sold,
3. whether the trade is opening or closing a position,
4. whether the transaction is a spread (discussed later), and
5. the desired price.

TYPES OF ORDERS

Many types of orders are acceptable for trading options, but not all are acceptable on all exchanges that trade options. Since regulations change, information regarding which order is valid for a given exchange is best supplied by the broker to the customer. The following orders are acceptable on all option exchanges, but certain electronic trading platforms will not take all types of orders:

Market Order. This is a simple order to buy or sell the option at the best possible price as soon as the order gets to the exchange.

Market Not Held Order. This type of order can only be used in situations where pit trading still exists—where a physical broker is handling the order in a crowd; it is not apropos to electronic trading. The customer who uses this type of order is giving the floor broker discretion in executing the order. The floor broker is not held responsible for the final outcome. For example, if a floor broker has a "market not held" order to buy, and he feels that the stock will "downtick" (decline in price) or that there is a surplus of sellers in the crowd, he may often hold off on the execution of the buy order, figuring that the price will decline shortly and that the order can then be executed at a more favorable price. In essence, the customer is giving the floor broker the right to use some judgment regarding the execution of the order. If the floor broker has an opinion and that opinion is correct, the customer will probably receive a better price than if he had used a regular market order. If the broker's opinion is wrong, however, the price of the execution may be worse than a regular market order.

Limit Order. The limit order is an order to buy or to sell at a specified price—the limit. It may be executed at a better price than the limit—a lower one for buyers and a higher one for sellers. However, if the limit is never reached, the order may never be executed.

In pit trading situations where a physical floor broker is handling the order sometimes a limit order may specify a discretionary margin for the floor broker. In other words, the order may read "Buy at 5 with dime discretion." This instruction enables the floor broker to execute the order at 5.10 if he feels that the market will never reach 5. Under no circumstances, however, can the order be executed at a price higher than 5.10.

Stop Order. This order is not always valid on all option exchanges. A stop order becomes a market order when the security trades at or through the price specified on the order. Buy stop orders are placed above the current market price, and sell stop orders are entered below the current market price. Such orders are used to either limit loss or protect a profit. For example, if a holder's option is selling for 3, a sell stop order for 2 is activated if the market drops down below the 2 level, whereupon the floor broker would execute the order as soon as possible. The customer, however, is not guaranteed that the trade will be exactly at 2. It should be noted that some exchanges will consider a stop order for an option to be elected (i.e., made live) if the bid price of the option falls to the limit—not necessarily the last sale of the option. This might mean that stop orders get elected in illiquid markets when the customer didn't really intend for it to. In general, one should be cautious about using stop orders with illiquid options.

Stop-Limit Order. This order becomes a limit order when the specified price is reached. Whereas the stop order has to be executed as soon as the stop price is reached, the stop-limit may or may not be filled, depending on market behavior. For example, suppose the option is currently trading at 3, and one has a stop-limit order to sell at 2. The market starts to drop, and trades at 2. Your stop-limit order becomes a limit order to sell at 2. However, if the option never trades at 2, but continues on down—1.90, 1.85, 1.80 and so forth—you will not sell your option. It cannot be sold at a price less than 2.

Good-Until-Canceled Order. A limit, stop, or stop-limit order may be designated "good until canceled." If the conditions for the order execution do not occur, the order remains valid for 6 months without renewal by the customer.

Customers using an on-line broker will not be able to enter "market not held" orders, and may not be able to use stop orders or good-until-canceled orders either, depending on the brokerage firm.

PROFITS AND PROFIT GRAPHS

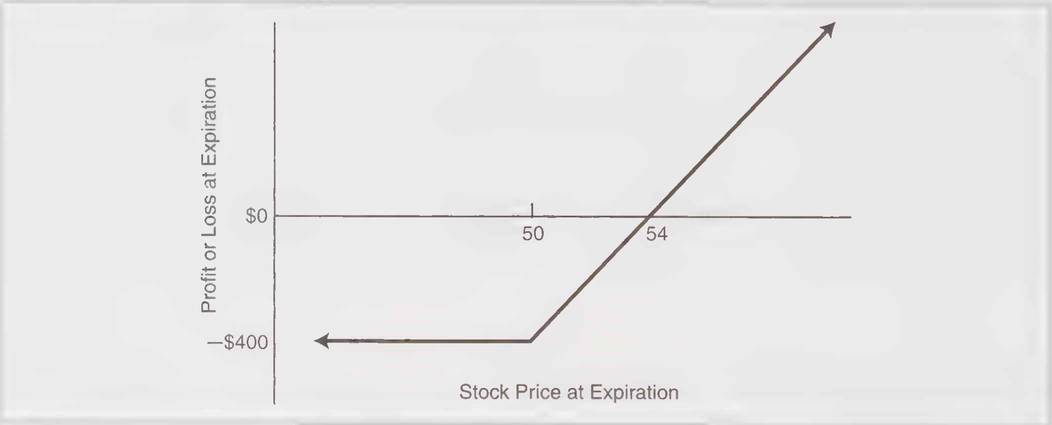
A visual presentation of the profit potential of any position is important to the over-all understanding and evaluation of it. In option trading, the many multi-security positions especially warrant strict analysis: stock versus options (as in covered or ratio writing) or options versus options (as in spreads). Some strategists prefer a table listing the outcomes of a particular strategy for the stock at various prices; others think the strategy is more clearly demonstrated by a graph. In the rest of the text, both a table and a graph will be presented for each new strategy discussed.

Example: A customer wishes to evaluate the purchase of a call option. The potential profits or losses of a purchase of an XYZ July 50 call at 4 can be arrayed in either a table or a graph of outcomes at expiration. Both Table 1-5 and Figure 1-5 depict the same information; the graph is merely the line representing the column marked "Profit or Loss" in the table. The

TABLE 1-5.
Potential profits and losses for an XYZ call purchase.

XYZ Price at Expiration	Call Price at Expiration	Profit or Loss
40	$ 0	– $ 400
45	0	– 400
50	0	– 400
55	5	+ 100
60	10	+ 600
70	20	+ 1,600

FIGURE 1-5.
Graph of potential profits for an XYZ call purchase.



vertical axis represents dollars of profit or loss, and the horizontal axis shows the stock price at expiration. In this case, the dollars of profit and the stock price are at the expiration date. Often, the strategist wants to determine what the potential profits and losses will be before expiration, rather than at the expiration date itself. Tables and graphs lend themselves well to the necessary analysis, as will be seen in detail in various places later on.

In practice, such an example is too simple to require a table or a graph—certainly not both—to evaluate the potential profits and losses of a simple call purchase held to expiration. However, as more complex strategies are discussed, these tools become ever more useful for quickly determining such things as when a position makes money and when it loses, or how fast one's risk increases at certain stock prices.